

BASSAM ARNAOUT

Sherbrooke, Canada

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CAREER SUMMARY

Data Scientist • Machine Learning Engineer • AI Developer

Results-driven Data Scientist with a Master's degree in Computer Science (Machine Learning, AI, Data Mining) and 10+ years of software development experience. Skilled in building predictive models, deep learning pipelines, and scalable data-driven solutions. Strong statistical foundation with expertise in Python, SQL, ML algorithms, and end-to-end development. Proven ability to deliver measurable business impact through analytical problem-solving.

- Enhanced ML model accuracy by 12% through advanced feature engineering and hyperparameter tuning.
- Automated data preprocessing workflows, reducing manual processing time by 40%.

CORE SKILLS

Machine Learning & Data Science

- Supervised & Unsupervised Learning
- Feature Engineering & Model Optimization
- Classification, Regression, Clustering
- Image Processing & Computer Vision
- Statistical Modeling & Predictive Analytics

Technical Skills

- Languages: Python, C#, C++, Java, SQL
- ML Tools: NumPy, Pandas, Scikit-learn, SciPy, XGBoost, MLP, SVM
- Deep Learning: Neural Networks, Hyperparameter Tuning
- Data Visualization: Tableau, Matplotlib
- Databases: MySQL, PostgreSQL, MS-SQL, MongoDB
- Cloud & Tools: Git, GitHub, REST APIs

Software Engineering

- Full SDLC, Architecture, API Development
- Legacy System Modernization
- High-performance backend systems

RELEVANT DATA SCIENCE & ML PROJECTS

Advanced Liver Fibrosis Prediction (ML & Deep Learning) (GitHub: bassamarnaout)

Developed machine learning and deep learning models to predict advanced liver fibrosis in chronic Hepatitis C patients.

- Algorithms: LR, LDA, KNN, NB, CART, SVM, XGBoost, MLP
- Tasks: Preprocessing, feature extraction, model training, hyperparameter tuning
- Tools: Python, Scikit-learn, TensorFlow
- Achieved strong classification accuracy across multiple models

Predictive Loan Default Model (GitHub: bassamarnaout)

Built a predictive model to estimate loan default probability using multiple classification algorithms.

- Used Logistic Regression, KNN, Gaussian NB, Decision Trees, SVM, Gradient Boosting, Extra Trees
- Conducted EDA, model comparison, and performance evaluation
- Improved prediction quality through feature selection

Image Analysis – Corrosion Detection: (GitHub: bassamarnaout)

Implemented GLCM-based texture extraction and WCCD algorithm to identify corroded regions in industrial images.

- Techniques: Texture Analysis, Segmentation
- Technologies: Python, OpenCV, NumPy

Image Segmentation Using Mean Shift: (GitHub: bassamarnaout)

Developed image segmentation algorithm using the mean-shift technique to separate color regions.

Social Media Usage Data Analysis (Tableau):

Performed data cleaning, EDA, and dashboard creation to visualize behavior across social platforms.

PROFESSIONAL EXPERIENCE

Senior Software Developer - Genatec Inc. - Montreal, Quebec

Nov 2020 – Present

- Built Python and C#/.NET applications integrated with Sage 300 ERP, automating financial processes.
- Modernized legacy software, improving performance and maintainability.
- Delivered scalable backend solutions and contributed to architecture decisions.
- Improved customer satisfaction through effective debugging and support.

Customer Service Representative - Nordia (Canada Post) - Sherbrooke, Quebec

August 2020 – Nov 2020

- Provided remote customer support with high communication standards.
- Awarded “Employee of the Week” 4 times for service excellence.

Freelancer – Web Developer - Econoplus Electromenagers - Sherbrooke, Quebec

May 2020 – July 2020

- Built an e-commerce platform and optimized UX for performance and navigation (Econoplus Électroménagers website).

IT Project Manager / Software Developer - Mob2link - Beirut, Lebanon

Jan 2016 – August 2018

- Led development of a Call Centre solution and the "Hayda Deal" mobile application.
- Managed product lifecycle, digital marketing, and technical delivery.
- Supervised the company Value Added Services portfolio based on market requirements & technology capabilities.
- Launched Intelligent Campaign Management System generating 9M SMS/month – Reduce Churn & Increase Usage-ARPU by 2.3%.

IT Manager / Software Developer - Asaig – Kuwait

Jan 2014 – Dec 2015

- Developed high-throughput SMS backend handling 555 SMS/sec using C#/.NET. Improved processing speed by 30%
- Led IT operations, infrastructure, and technical teams

EDUCATION

Master of Science (M.Sc.) in Computer Science - Bishop's University - Sherbrooke, Canada.

2018 - 2020

Specialization: Machine Learning, Data Mining, Artificial Intelligence, Deep Learning, Neural Network.

Bachelor (BE) in Electrical & Electronic Engineering - AUB (American University of Beirut) – Lebanon

Focus: Computer & Communication Engineering Department CCE)

CERTIFICATIONS & TRAINING

- Microsoft Certified Professional (MCP)
- MCSD.NET Training (New Horizon)
- PMP 35-hour training (Lifelong Learning)

LANGUAGES

- English: Fluent
- French: Intermediate